Piramu-Mahalingam / DS-Course-Assignment-1

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DS-Course-Assignment-1 / Mini_project_1.ipynb

Piramu-Mahalingam Created using Colab 265e2e6 · 1 hour ago History

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In [ ]:

```
#Library Management System-Mini Project
```

In [20]:

```
#Step 1: Basic Structure
print('Welcome for the Library Management System')
library={"Python Basics":{"author":'Test',"Available":True},
         "Data Science":{"author":'Test1',"Available":False}}

#view books details
def view_books(library):
  for a,b,in library.items():
    print(a,b)

#Step 2: Adding Books
def add_books():

    while True:
        book_name = input("Enter the title of the book (e.g., 'Python Basics'): ")

        # Ensure book name is unique in the library
        if book_name in library:
            print("This book already exists in the library.")
        else:
            # Input nested dictionary attributes
            author = input(f"Enter the author of '{book_name}': ")
            available = input(f"Is '{book_name}' available? (True/False): ").strip().lower()
            available = available == 'true'  # Convert to boolean

            # Add book to the library with nested dictionary for attributes
            library[book_name] = {
                "author": author,
                "Available": available
            }

        # Ask if the user wants to input another book
        more = input("Do you want to add another book? (yes/no): ").strip().lower()
        if more != 'yes':
            break

    view_books(library)

#Step 3: Issuing Books
book={"Data Science":{"author":"Test1","Available":True,"Issued_to":None},
      "Machine Learning":{"author":"Test2","Available":False, "Issued_to":"Bob"}}
def issue_book(title, person_name):
    if title in book:
        b2 = book[title]
        if b2["Available"]:
            b2["Available"] = False
```

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            b2["Issued_to"] = person_name
            print(f"b2 '{title}' has been issued to {person_name}.")
        else:
            print(f"Sorry, '{title}' is already issued to {b2['Issued_to']}.")
    else:
        print(f"book '{title}' does not exist.")

#Step 4: Returning Books
def return_book(title):
    # Check if the book exists and is issued
    if title in book:
        b3 = book[title]
        if not b3["Available"]:
            print(f"Book '{title}' has been returned by {b3['Issued_to']}.")
            b3["Available"] = True
            b3["Issued_to"] = None
        else:
            print(f"Book '{title}' was not issued, so it cannot be returned.")
    else:
        print(f"Book '{title}' does not exist.")

#Step 5: Login System
def login_system():
    # Predefined credentials (username: password)
    credentials = {"admin": "admin123", "librarian": "lib123"}

    # Allow up to 3 login attempts
    attempt = 3

    while attempt > 0:
        # Prompt user for username and password
        username = input("Enter your username: ")
        password = input("Enter your password: ")

        # Check if the username exists and if the password matches
        if username in credentials and credentials[username] == password:
            print("Login successful! Welcome to the system.")
            break
        else:
            attempt -= 1
            if attempt > 0:
                print(f"Invalid credentials. You have {attempt} attempts left.")
            else:
                print("Too many failed attempts.")
                break

#Step 6: View Issued Books
def display_status(title):
  if title in book:
    b1=book[title]
    if b1["Available"]:
      print("book is available",b1)
    else:
```

```
      print(f"book:{title} by {b1['author']} is issued to {b1['Issued_to']}.")
  else:
    print("invalid book title")

#Step 7: Main Menu
login_system()
while True:
            print("Menu")
            print("1. View Books ")
            print("2. Add a Book ")
            print("3. Issue a Book ")
            print("4. Return a Book ")
            print("5. View Issued Books ")
            print("6. Exit ")
            choice=input("Enter the option")

            if(choice=="1"):
              view_books(library)
            elif(choice=="2"):
              add_books()
            elif(choice=='3'):
              title=input("Enter book title")
              person_name=input("Enter the person name")
              issue_book(title,person_name)
            elif(choice=='4'):
              title=input("Enter book title")
              return_book(title)
            elif(choice=='5'):
              title=input("Enter book title")
              display_status(title)
            elif(choice=="6"):
              print("Good bye")
              break
            else:
              print("invalid choice")
```

```
Welcome for the Library Management System
Enter your username: admin123
Enter your password: admin
Invalid credentials. You have 2 attempts left.
Enter your username: admin
Enter your password: admin123
Login successful! Welcome to the system.
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option1
Python Basics {'author': 'Test', 'Available': True}
Data Science {'author': 'Test1', 'Available': False}
Menu
```

```
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option2
Enter the title of the book (e.g., 'Python Basics'): Machine Learning
Enter the author of 'Machine Learning': Test2
Is 'Machine Learning' available? (True/False): False
Do you want to add another book? (yes/no): no
Python Basics {'author': 'Test', 'Available': True}
Data Science {'author': 'Test1', 'Available': False}
Machine Learning {'author': 'Test2', 'Available': False}
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option3
Enter book titleData Science
Enter the person nameAda
b2 'Data Science' has been issued to Ada.
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option3
Enter book titleMachine Learning
Enter the person nameGanesh
Sorry, 'Machine Learning' is already issued to Bob.
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option5
Enter book titleData Science
book:Data Science by Test1 is issued to Ada.
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option4
Enter book titleData Science
Book 'Data Science' has been returned by Ada.
```

```
Book 'Data Science' has been returned by Ada.
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option5
Enter book titleData Science
book is available {'author': 'Test1', 'Available': True, 'Issued_to': None}
Menu
1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit
Enter the option6
Good bye
```